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Theoretical Physics

Quantum Many Body Theory

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Introduction

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1 | Weakly Interacting Bose Gas

1.1 | The Weak Interaction Assumption

The weakly interacting Bose gas serves as a foundational paradigm in quantum many-body theory. We consider a system of N identical bosons of mass m confined to a box of volume $V = L^3$ with periodic boundary conditions.

The interparticle interaction is characterized by a two-body potential $U(\mathbf{r})$, which we assume to be weak. The constituents of this gas obey Bose-Einstein statistics and are treated as non-relativistic elementary particles, meaning that their internal degrees of freedom are not activated at the relevant energy scales, and the total particle number is strictly fixed (precluding, for example, a gas of photons).

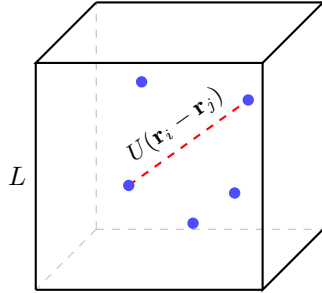


Figure 1.1: Schematic representation of a system of N interacting bosons confined within a cubic box of volume $V = L^3$. The periodic boundary conditions simulate a bulk system. The dashed red line illustrates the two-body interaction potential $U(\mathbf{r}_i - \mathbf{r}_j)$ acting strictly between pairs of particles.

Typical experimental realizations involve dilute gases of bosonic atoms, meaning atoms possessing an even total number of constituent fermions (protons, neutrons, and electrons), such as Sodium (^{23}Na), Rubidium (^{87}Rb), Potassium (^{41}K), Cesium (^{133}Ce), and Lithium (^7Li). Historically, this theoretical framework was first motivated by efforts to explain the superfluid behavior of Helium-4 below 2.17K, despite the fact that liquid ^4He actually resides in a strongly interacting regime.

The Hamiltonian governing this system of N interacting identical bosons is given in first quantization by

$$\hat{H} = \sum_{i=1}^N \frac{\hat{\mathbf{p}}_i^2}{2m} + \frac{1}{2} \sum_{\substack{i,j \\ i \neq j}} U(\hat{\mathbf{r}}_i - \hat{\mathbf{r}}_j),$$

where $\hat{\mathbf{p}}_i$ is the momentum operator of the i -th particle, and $U(\hat{\mathbf{r}}_i - \hat{\mathbf{r}}_j)$ represents the interaction energy between the i -th and j -th particles.

For a system comprising a macroscopic number of identical bosons, it is highly advantageous to

transition to the second quantization formalism.¹ We introduce bosonic creation and annihilation operators associated with the momentum eigenfunctions of a single particle. For a box with periodic boundary conditions, these single-particle states are quantized plane waves:

$$\phi_{\mathbf{k}}(\mathbf{r}) = \frac{e^{i\mathbf{k}\cdot\mathbf{r}}}{\sqrt{V}},$$

where the allowed wave vectors $\mathbf{k}$ take the discrete values

$$\mathbf{k} = \frac{\pi}{L}(n_x, n_y, n_z), \quad n_x, n_y, n_z \in \{0, \pm 1, \pm 2, \dots\}.$$

In this momentum basis, the many-body Hamiltonian becomes

$$\hat{H} = \sum_{\mathbf{k}} \epsilon_k \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}} + \frac{1}{2V} \sum_{\mathbf{k}, \mathbf{k}', \mathbf{q}} U(\mathbf{q}) \hat{a}_{\mathbf{k}+\mathbf{q}}^\dagger \hat{a}_{\mathbf{k}'-\mathbf{q}}^\dagger \hat{a}_{\mathbf{k}'} \hat{a}_{\mathbf{k}}. \quad (1.1.1)$$

Here, ϵ_k denotes the non-interacting energy associated with the wave vector $\mathbf{k}$ (which depends exclusively on its magnitude k), given by the free-particle dispersion

$$\epsilon_k = \frac{\hbar^2 k^2}{2m},$$

and $U(\mathbf{q})$ is the Fourier transform of the real-space two-body potential $U(\hat{\mathbf{r}}_1 - \hat{\mathbf{r}}_2)$. The operators $\hat{a}_{\mathbf{k}}^\dagger$ and $\hat{a}_{\mathbf{k}}$ represent the creation and destruction operators for the corresponding momentum eigenstates.

In the idealized limit where interactions are completely absent, the ground state of the system is trivial: all N bosons condense into the lowest-energy single-particle state featuring zero momentum ($\mathbf{k} = 0$). This macroscopic formation of a condensate is mathematically expressed by acting N times with the zero-momentum creation operator $\hat{a}_0^\dagger$ on the vacuum state $|0\rangle$ (incorporating the standard $1/\sqrt{N!}$ normalization factor for formal completeness):

$$|\psi_0^{(0)}\rangle = \frac{(\hat{a}_0^\dagger)^N}{\sqrt{N!}} |0\rangle. \quad (1.1.2)$$

Remark. *This clustering phenomenon stands in stark contrast to the behavior of a gas of fermions. Governed by the Pauli exclusion principle, fermions are strictly forbidden from occupying identical quantum states. Consequently, even at zero temperature, non-interacting fermions must sequentially fill all available momentum states up to a maximum threshold known as the Fermi energy, precluding any analogous zero-momentum condensation.*

We now introduce the principal assumption of our approach: the interparticle interaction is sufficiently weak. Given this, we anticipate that the true, interacting ground state remains a modest perturbation of the unperturbed condensate state $|\psi_0^{(0)}\rangle$. Therefore, even in the presence of interactions, the vast majority of the bosons will remain in the zero-momentum single-particle state. This physical condition is formalized by the inequality

$$\frac{N - N_0}{N} \ll 1, \quad (1.1.3)$$

where $N_0 = \langle \hat{a}_0^\dagger \hat{a}_0 \rangle$ defines the thermodynamic average number of bosons residing in the condensate. The quantity $N - N_0$ thus counts the number of depleted bosons that have been excited out of the condensate into states with $\mathbf{k} \neq 0$. Due to the weakness of the interaction, these non-zero

¹For a very clear introduction to the second quantization formalism for many-particle systems, see: H. Tasaki, <https://arxiv.org/pdf/1812.10732v5>.

momentum states are only microscopically occupied, meaning their occupation numbers scale as $\langle \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}} \rangle = \mathcal{O}(1)$. Crucially, the population of these excited states does not scale proportionally with the macroscopic total particle number N .

We leverage this assumption to implement a fundamental approximation on the interaction term of the Hamiltonian (1.1.1). Because operators acting on the condensate ($\hat{a}_0$ and $\hat{a}_0^\dagger$) scale macroscopically with $\sqrt{N_0} \approx \sqrt{N}$, we can safely neglect all interaction terms containing more than two operators equipped with non-zero momentum ($\hat{a}_{\mathbf{k}}$ or $\hat{a}_{\mathbf{k}}^\dagger$ for $\mathbf{k} \neq 0$). (Note that terms containing exactly three zero-momentum operators are inherently forbidden by momentum conservation requirements).

The most substantial and relevant term that survives this truncation is the one containing exclusively operators acting on the condensate ($\mathbf{q} = 0$ and $\mathbf{k} = \mathbf{k}' = 0$):

$$\frac{1}{2V} U(\mathbf{q} = 0) \hat{a}_0^\dagger \hat{a}_0^\dagger \hat{a}_0 \hat{a}_0 = \frac{1}{2V} U(\mathbf{q} = 0) (\hat{N}_0^2 - \hat{N}_0).$$

In the thermodynamic limit, $\hat{N}_0$ diverges linearly with N . Thus, we can completely disregard the trailing $\hat{N}_0$ term in the above expression, as it scales infinitely slower than the dominant quadratic term $\hat{N}_0^2 \sim N^2$.

To systematically express this leading-order contribution in terms of the conserved total particle number, we recognize that the number of bosons in the ground state can be written as the difference between the total number operator $\hat{N}$ and the sum of occupation numbers for all excited states:

$$\hat{N}_0 = \hat{N} - \sum_{\mathbf{k} \neq 0} \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}},$$

where the primed summation specifically denotes a sum over all allowed momenta excluding zero. Letting $U_0 \equiv U(\mathbf{q} = 0)$, we substitute this relation back into our dominant interaction term. Expanding the square and purposefully discarding the resulting quartic term (which contains four out-of-condensate operators and is therefore negligible under our approximation), we obtain

$$\frac{1}{2V} U_0 (\hat{N}_0)^2 = \frac{1}{2V} U_0 \left(\hat{N} - \sum_{\mathbf{k} \neq 0} \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}} \right)^2 \approx \frac{1}{2V} U_0 \left(\hat{N}^2 - 2\hat{N} \sum_{\mathbf{k} \neq 0} \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}} \right).$$

Through this procedure, we successfully reduce the pure zero-momentum interaction into a quadratic form that couples the macroscopic total number N with the single-particle excitations.

Quadratic Interaction Terms and the Short-Range Assumption

Besides the dominant term where all operators are in the condensate state, we must evaluate the terms from the interaction Hamiltonian that contain exactly two operators with momentum different from zero. Before explicitly listing them, we introduce a crucial physical approximation: we replace the momentum-dependent interaction potential $U(\mathbf{q})$ with a simple constant, $U(0)$.

This approximation is mathematically justified if the real-space potential is effectively “short-ranged”. For the Fourier transform $U(\mathbf{q}) = \int d^3\mathbf{r} U(\mathbf{r}) e^{-i\mathbf{q} \cdot \mathbf{r}}$ to be well-behaved and approximately constant for small momenta, its zero-momentum limit $U(0) = \int d^3\mathbf{r} U(\mathbf{r})$ must be finite. In three spatial dimensions, the spherical integration measure introduces a factor of r^2 . Consequently, for the integral $\int dr r^2 U(r)$ to converge at large interparticle distances, the potential $U(r)$ must decay strictly faster than $1/r^3$.

This convergence criterion dictates which physical systems can be described by this simplified model. In dilute atomic gases, van der Waals forces serve as an excellent example of a valid short-ranged potential. In the standard Lennard-Jones model, the long-range attractive tail scales as $1/r^6$. Since $6 > 3$, it comfortably satisfies the convergence requirement. In contrast, dipole-dipole interactions scale exactly as $1/r^3$. This is a tricky marginal case where the spatial integral diverges logarithmically, meaning dipole interactions are mathematically long-ranged in 3D and cannot be treated with this simple approximation. Assuming our potential decays fast enough, we can safely set $U(\mathbf{q}) \approx U(\mathbf{q} = 0) \equiv U_0$, making all subsequent calculations significantly simpler.

Remark. *As an analogous example of effective short-range interactions from high-energy physics, the strong nuclear force is also effectively short-ranged (decaying exponentially, much faster than $1/r^3$) because its mediators, pions, are massive.*

With the assumption $U(\mathbf{q}) \approx U_0$ established, we retain the following six terms containing exactly two non-zero momentum operators:

1. If $\mathbf{k} + \mathbf{q} = 0$ and $\mathbf{k}' - \mathbf{q} = 0$, we have that $\mathbf{k} = -\mathbf{q}$ and $\mathbf{k}' = \mathbf{q}$, and so we remain with

$$\frac{1}{2V} U_0 a_0^\dagger a_0^\dagger \sum_{\mathbf{q}}' a_{\mathbf{q}} a_{-\mathbf{q}},$$

where the sum $\sum_{\mathbf{q}}'$ is over all possible momenta $\mathbf{q} \neq 0$ because the zero case has been already taken into account in the previous purely macroscopic term.

2. If $\mathbf{k} + \mathbf{q} = 0$ and $\mathbf{k}' = 0$, using again the short-range assumption, the term takes the following form:

$$\frac{1}{2V} U_0 \sum_{\mathbf{q}}' a_0^\dagger a_{-\mathbf{q}}^\dagger a_0 a_{-\mathbf{q}} = \frac{1}{2V} U_0 a_0^\dagger a_0 \sum_{\mathbf{q}}' a_{-\mathbf{q}}^\dagger a_{-\mathbf{q}}.$$

3. If $\mathbf{k} + \mathbf{q} = 0$ and $\mathbf{k} = 0$, and so $\mathbf{q} = 0$, we get

$$\frac{1}{2V} U_0 \sum_{\mathbf{k}'}' a_0^\dagger a_{\mathbf{k}'}^\dagger a_{\mathbf{k}'} a_0 = \frac{1}{2V} U_0 a_0^\dagger a_0 \sum_{\mathbf{k}'}' a_{\mathbf{k}'}^\dagger a_{\mathbf{k}'}.$$

4. If $\mathbf{k}' - \mathbf{q} = 0$ and $\mathbf{k}' = 0$, and so $\mathbf{q} = 0$, we get

$$\frac{1}{2V} U_0 \sum_{\mathbf{k}}' a_{\mathbf{k}}^\dagger a_0^\dagger a_0 a_{\mathbf{k}} = \frac{1}{2V} U_0 a_0^\dagger a_0 \sum_{\mathbf{k}}' a_{\mathbf{k}}^\dagger a_{\mathbf{k}}.$$

5. If $\mathbf{k}' - \mathbf{q} = 0$ and $\mathbf{k} = 0$, with the short-range assumption the term takes the following form:

$$\frac{1}{2V} U_0 \sum_{\mathbf{q}}' a_{\mathbf{q}}^\dagger a_0^\dagger a_{\mathbf{q}} a_0 = \frac{1}{2V} U_0 a_0^\dagger a_0 \sum_{\mathbf{q}}' a_{\mathbf{q}}^\dagger a_{\mathbf{q}}.$$

6. If $\mathbf{k}' = 0$ and $\mathbf{k} = 0$, with the short-range assumption the term takes the following form:

$$\frac{1}{2V} U_0 \sum_{\mathbf{q}}' a_{\mathbf{q}}^\dagger a_{-\mathbf{q}}^\dagger a_0 a_0 = \frac{1}{2V} U_0 a_0 a_0 \sum_{\mathbf{q}}' a_{\mathbf{q}}^\dagger a_{-\mathbf{q}}^\dagger.$$

Condensate Mean-Field Approximation and the Bogoliubov Hamiltonian

A further crucial step is the replacement of the operators a_0 and $a_0^\dagger$ with numbers. Because the non-interacting ground state exhibits macroscopic occupation, its expectation value is $\langle a_0^\dagger a_0 \rangle = N_0 \approx N$. Therefore, we can confidently approximate $a_0 \sim \sqrt{N}$ and $a_0^\dagger \sim \sqrt{N}$ across all the previously derived terms of the approximate Hamiltonian that contain two condensate operators multiplied by two out-of-condensate operators.²

This mean-field approximation is formally justified by inspecting the commutator $[a_0, a_0^\dagger] = 1$ and comparing its magnitude to the result of acting with these operators on the non-interacting N -particle ground state $|\psi_0^{(0)}(N)\rangle$. Specifically, one finds:

$$a_0 |\psi_0^{(0)}(N)\rangle = \sqrt{N} |\psi_0^{(0)}(N-1)\rangle \sim \sqrt{N} |\psi_0^{(0)}(N)\rangle,$$

and

$$a_0^\dagger |\psi_0^{(0)}(N)\rangle = \sqrt{N+1} |\psi_0^{(0)}(N+1)\rangle \sim \sqrt{N} |\psi_0^{(0)}(N)\rangle,$$

where we have safely neglected 1 relative to the macroscopically large number N in these expressions. Because of this scaling, the value of the commutator is entirely negligible, and the operators can be effectively treated as commuting c-numbers.

Furthermore, during this systematic expansion, it is important to recognize why linear terms in $a_{\mathbf{k}}$ or $a_{\mathbf{k}}^\dagger$ do not appear. The terms which cancel out are precisely the ones that contain a single $\hat{a}_0$ or $\hat{a}_0^\dagger$ in the interaction term; if they did not cancel, they would have contributed to an unphysical renormalization of the free term of the Hamiltonian.

By substituting these numerical expectations back into our aggregated interaction terms, the full Hamiltonian becomes:

TODO: expand computations.

$$H = \sum_{\mathbf{k}}' (\epsilon_{\mathbf{k}} + nU_0) a_{\mathbf{k}}^\dagger a_{\mathbf{k}} + \frac{nU_0}{2} \sum_{\mathbf{k}}' (a_{-\mathbf{k}}^\dagger a_{\mathbf{k}}^\dagger + a_{\mathbf{k}} a_{-\mathbf{k}}) + \frac{1}{2} V n^2 U_0, \quad (1.1.4)$$

where $n = N/V$ is the particle density.

Note that the second sum in the above equation contains the “anomalous” terms $a_{-\mathbf{k}}^\dagger a_{\mathbf{k}}^\dagger$ and $a_{\mathbf{k}} a_{-\mathbf{k}}$. Because the Hamiltonian now pairs creation operators together (and annihilation operators together) rather than consisting purely of standard number operators, it is not diagonal. To resolve this and find the true elementary excitations, one must employ the Bogoliubov transformation to diagonalize the Hamiltonian.

As a final remark, we note that if one were to carry out this derivation without invoking the constant potential approximation $U(\mathbf{q}) \simeq U_0$, the resulting Hamiltonian would naturally retain the momentum dependence of the interaction:

$$H = \sum_{\mathbf{k}}' (\epsilon_{\mathbf{k}} + nU(k)) a_{\mathbf{k}}^\dagger a_{\mathbf{k}} + \frac{n}{2} \sum_{\mathbf{k}}' U(k) (a_{-\mathbf{k}}^\dagger a_{\mathbf{k}}^\dagger + a_{\mathbf{k}} a_{-\mathbf{k}}) + \frac{1}{2} V n^2 U_0.$$

²This can be pretty disturbing, since the expectation value of a single ladder operator is usually zero; but, as we will see later, we are not using eigenstates of number operator, instead superpositions of those (similar to coherent states). Indeed, even if the hamiltonian conserves the number of particle, the ground state description does not really rely on a fixed occupation of bosons. We will stress this point later on.

Summary of Assumptions and Approximations

Before proceeding to diagonalize the Hamiltonian, it is instructive to summarize the hierarchy of physical assumptions and mathematical approximations that have systematically reduced the complex many-body problem to the quadratic Bogoliubov Hamiltonian (1.1.4):

- **Non-Relativistic Elementary Bosons:** We assume the particles are massive, non-relativistic, and constrained to energy scales where internal degrees of freedom are completely frozen. *Consequence:* The total particle number N is strictly conserved (unlike in a photon gas), and the system is well-described by standard Galilean kinetic energy and a static scalar two-body potential.
- **Weak Interaction Hypothesis:** The interparticle potential is assumed to be sufficiently weak such that the true interacting ground state is only a modest perturbation of the ideal Bose-Einstein condensate. *Consequence:* The fraction of depleted particles is vanishingly small $((N - N_0)/N \ll 1)$. This justifies truncating the interaction Hamiltonian to discard all terms containing more than two out-of-condensate operators ($\mathbf{k} \neq 0$).
- **Short-Range Potential:** The two-body interaction $U(\mathbf{r})$ is assumed to decay strictly faster than $1/r^3$ at large interparticle distances. *Consequence:* The Fourier transform of the potential converges and varies slowly at small momenta, allowing us to approximate $U(\mathbf{q}) \approx U(\mathbf{q} = 0) \equiv U_0$. This dramatically simplifies the scattering amplitudes in the dilute gas regime.
- **Macroscopic Condensate Occupation (Mean-Field Limit):** The zero-momentum state is macroscopically occupied ($N_0 \approx N \gg 1$), meaning the fundamental commutator $[\hat{a}_0, \hat{a}_0^\dagger] = 1$ is negligible compared to the absolute particle number. *Consequence:* The macroscopic creation and annihilation operators $\hat{a}_0^\dagger$ and $\hat{a}_0$ can be treated as commuting c-numbers and replaced by $\sqrt{N}$. This breaks the $U(1)$ phase symmetry of the Hamiltonian and reduces the complicated quartic interaction term into a tractable quadratic form featuring anomalous pairing terms.

1.1.1 | Bogoliubov Transformations

We wish to diagonalize the Hamiltonian of the weakly interacting Bose gas by introducing new bosonic annihilation and creation operators, $\alpha_{\mathbf{k}}$ and $\alpha_{\mathbf{k}}^\dagger$, known as Bogoliubov operators. We need to construct a Hamiltonian which, in the end, depends only on a diagonal number operator of the form $\alpha_{\mathbf{k}}^\dagger \alpha_{\mathbf{k}}$, yielding a target structure like:

$$H = \sum_{\mathbf{k}} E_{\mathbf{k}} \alpha_{\mathbf{k}}^\dagger \alpha_{\mathbf{k}} + \text{const},$$

for some energy dispersion $E_{\mathbf{k}}$. To eliminate the anomalous terms that mix creation and annihilation operators in our previous Hamiltonian, we can perform a Bogoliubov transformation, which is a linear transformation mapping the original particle operators to the new quasiparticle operators. We start with the following standard ansatz for the Bogoliubov transformation:

$$\begin{cases} \alpha_{\mathbf{k}} = u_{\mathbf{k}} a_{\mathbf{k}} + v_{\mathbf{k}} a_{-\mathbf{k}}^\dagger, \\ \alpha_{\mathbf{k}}^\dagger = v_{\mathbf{k}}^* a_{\mathbf{k}}^\dagger + u_{\mathbf{k}}^* a_{-\mathbf{k}}, \end{cases} \quad (1.1.5)$$

for $\mathbf{k} \neq 0$, where u_k and v_k are real coefficients. We now assume that these coefficients are real for simplicity, but they could in principle be complex. These coefficients are defined to depend strictly on the momentum magnitude $k = |\mathbf{k}|$, ensuring that rotational invariance is preserved in momentum space.

Because these newly defined operators must describe valid bosonic excitations, they are strictly required to satisfy the standard canonical bosonic commutation relations. We can verify this by evaluating the commutator between two annihilation operators:

$$\begin{aligned} [\alpha_{\mathbf{k}}, \alpha_{\mathbf{k}'}] &= [u_k a_{\mathbf{k}} + v_k a_{-\mathbf{k}}^\dagger, u_{k'} a_{\mathbf{k}'} + v_{k'} a_{-\mathbf{k}'}^\dagger] \\ &= u_k v_{k'} [a_{\mathbf{k}}, a_{-\mathbf{k}'}^\dagger] + v_k u_{k'} [a_{-\mathbf{k}}^\dagger, a_{\mathbf{k}'}] \\ &= u_k v_{k'} \delta_{\mathbf{k}, -\mathbf{k}'} - v_k u_{k'} \delta_{-\mathbf{k}, \mathbf{k}'} = 0, \end{aligned}$$

which vanishes as expected. Next, we evaluate the critical commutator between a creation and an annihilation operator:

$$\begin{aligned} [\alpha_{\mathbf{k}}, \alpha_{\mathbf{k}'}^\dagger] &= [u_k a_{\mathbf{k}} + v_k a_{-\mathbf{k}}^\dagger, u_{k'} a_{\mathbf{k}'}^\dagger + v_{k'} a_{-\mathbf{k}'}] \\ &= u_k u_{k'} [a_{\mathbf{k}}, a_{\mathbf{k}'}^\dagger] + v_k v_{k'} [a_{-\mathbf{k}}^\dagger, a_{-\mathbf{k}'}] \\ &= (u_k^2 - v_k^2) \delta_{\mathbf{k}, \mathbf{k}'}. \end{aligned}$$

To enforce the correct commutation relations, we must impose a hyperbolic normalization condition on the coefficients:

$$|u_k|^2 - |v_k|^2 = 1. \quad (1.1.6)$$

Provided this condition holds, the transformation linking the old and new operators can be easily inverted. We can express the linear mapping and its Hermitian conjugate in matrix notation:

$$\begin{pmatrix} \alpha_{\mathbf{k}} \\ \alpha_{-\mathbf{k}}^\dagger \end{pmatrix} = \begin{pmatrix} u_k & v_k \\ v_k & u_k \end{pmatrix} \begin{pmatrix} a_{\mathbf{k}} \\ a_{-\mathbf{k}}^\dagger \end{pmatrix}.$$

Because the determinant of the transformation matrix is exactly $u_k^2 - v_k^2 = 1$, the matrix is trivially invertible:

$$\begin{pmatrix} a_{\mathbf{k}} \\ a_{-\mathbf{k}}^\dagger \end{pmatrix} = \begin{pmatrix} u_k & -v_k \\ -v_k & u_k \end{pmatrix} \begin{pmatrix} \alpha_{\mathbf{k}} \\ \alpha_{-\mathbf{k}}^\dagger \end{pmatrix},$$

which explicitly yields the inverted transformation:

$$\begin{cases} a_{\mathbf{k}} = u_k \alpha_{\mathbf{k}} - v_k \alpha_{-\mathbf{k}}^\dagger, \\ a_{\mathbf{k}}^\dagger = u_k \alpha_{\mathbf{k}}^\dagger - v_k \alpha_{-\mathbf{k}}. \end{cases}$$

We now proceed with the formal diagonalization of the Hamiltonian by expressing it entirely in terms of the new Bogoliubov operators:

$$\begin{aligned} H &= \sum_{\mathbf{k}}' [(\epsilon_k + nU_0)v_k^2 - nU_0 u_k v_k] + \frac{1}{2} V n^2 U_0 \\ &\quad + \sum_{\mathbf{k}}' [(\epsilon_k + nU_0)(u_k^2 + v_k^2) - 2nU_0 u_k v_k] \alpha_{\mathbf{k}}^\dagger \alpha_{\mathbf{k}} \\ &\quad + \sum_{\mathbf{k}}' \left[\frac{nU_0}{2} (u_k^2 + v_k^2) - u_k v_k (\epsilon_k + nU_0) \right] (\alpha_{\mathbf{k}}^\dagger \alpha_{-\mathbf{k}}^\dagger + \alpha_{-\mathbf{k}} \alpha_{\mathbf{k}}). \end{aligned}$$

It may initially appear that we have not made significant progress, as the final line still contains non-diagonal anomalous terms. However, we now have the mathematical freedom to choose arbitrary coefficients that eliminate these specific terms. By selecting an appropriate v_k (which

simultaneously determines u_k via the constraint $u_k^2 = 1 + v_k^2$, we can force the bracketed prefactor of the off-diagonal terms to vanish identically:

$$\frac{nU_0}{2}(u_k^2 + v_k^2) - u_k v_k (\epsilon_k + nU_0) = 0.$$

Through simple algebraic manipulation and applying our normalization condition, this requirement simplifies to a second-order polynomial equation for v_k^2 :

$$4E_k^2 v_k^4 + 4E_k^2 v_k^2 - (nU_0)^2 = 0,$$

where we have suggestively defined the factor E_k^2 as:

$$E_k^2 = (\epsilon_k + nU_0)^2 - (nU_0)^2. \quad (1.1.7)$$

Solving this equation for v_k^2 directly yields the functional form of the transformation coefficients:

$$v_k^2 = \frac{1}{2} \left(\frac{\epsilon_k + nU_0}{E_k} - 1 \right),$$

and consequently,

$$u_k^2 = \frac{1}{2} \left(\frac{\epsilon_k + nU_0}{E_k} + 1 \right),$$

which implies for the cross-term:

$$u_k v_k = \frac{nU_0}{2E_k}.$$

By inserting these rigorously derived coefficients back into our expanded operator expression, the off-diagonal terms strictly cancel, yielding the final form of the diagonalized Hamiltonian:

$$H = \sum_{\mathbf{k}}' E_k \alpha_{\mathbf{k}}^\dagger \alpha_{\mathbf{k}} + \mathcal{E}_0, \quad (1.1.8)$$

where $\mathcal{E}_0$ is a macroscopic constant energy shift defined by:

$$\mathcal{E}_0 = \frac{1}{2} V n^2 U_0 - \frac{1}{2} \sum_{\mathbf{k}}' (\epsilon_k + nU_0 - E_k).$$

The quantity $E_k = [(\epsilon_k + nU_0)^2 - (nU_0)^2]^{1/2} > 0$ represents the fundamental quasiparticle excitation spectrum of the system.

The Non-Interacting Limit

In the non-interacting limit, characterized by a vanishing interaction strength $U_0 \rightarrow 0$, it is instructive to verify that our diagonalized theory smoothly recovers the foundational physics of the ideal Bose gas.

First, we examine the Bogoliubov quasiparticle excitation spectrum E_k . By taking the limit of the dispersion relation, we find:

$$E_k = \lim_{U_0 \rightarrow 0} \sqrt{(\epsilon_k + nU_0)^2 - (nU_0)^2} = \sqrt{\epsilon_k^2} = \epsilon_k.$$

As physically expected, the quasiparticle spectrum trivially reduces to the free-particle kinetic energy $\epsilon_k = \hbar^2 k^2 / (2m)$.

Next, we evaluate the asymptotic behavior of the Bogoliubov transformation coefficients v_k^2 and u_k^2 . Substituting $E_k \rightarrow \epsilon_k$ and $U_0 \rightarrow 0$, we obtain:

$$v_k^2 = \lim_{U_0 \rightarrow 0} \frac{1}{2} \left(\frac{\epsilon_k + nU_0}{E_k} - 1 \right) = \frac{1}{2}(1 - 1) = 0,$$

and correspondingly,

$$u_k^2 = \lim_{U_0 \rightarrow 0} \frac{1}{2} \left(\frac{\epsilon_k + nU_0}{E_k} + 1 \right) = \frac{1}{2}(1 + 1) = 1.$$

This implies that $u_k \rightarrow 1$ and $v_k \rightarrow 0$. Consequently, the Bogoliubov quasiparticle operators $\alpha_{\mathbf{k}} = u_k a_{\mathbf{k}} + v_k a_{-\mathbf{k}}^\dagger$ reduce identically to the bare particle operators $a_{\mathbf{k}}$. In the absence of interactions, the elementary excitations of the system are simply the original non-interacting bosons themselves, and the quantum depletion of the condensate drops to zero.

Finally, we analyze the macroscopic constant energy shift $\mathcal{E}_0$. The purely classical mean-field contribution $\frac{1}{2} V n^2 U_0$ obviously vanishes. For the quantum fluctuation sum, we note that $E_k \approx \epsilon_k + nU_0 - \frac{(nU_0)^2}{2\epsilon_k}$ for small U_0 , meaning the summand $(\epsilon_k + nU_0 - E_k) \sim \mathcal{O}(U_0^2)$. Therefore, the entire shift vanishes:

$$\lim_{U_0 \rightarrow 0} \mathcal{E}_0 = 0.$$

The ground state energy of the system correctly defaults to zero, recovering the trivial result for an ideal Bose-Einstein condensate where all particles occupy the lowest available energy level with zero kinetic energy.

Ground State

With the Hamiltonian successfully diagonalized, we first seek its ground state, denoted $|\psi_0\rangle$. This state is mathematically defined as the vacuum of the new Bogoliubov annihilation operators $\alpha_{\mathbf{k}}$ for all non-zero momenta:

$$\alpha_{\mathbf{k}} |\psi_0\rangle = 0, \quad \forall \mathbf{k} \neq 0,$$

since any mode added to the vacuum would necessarily increase the energy of the system.

The corresponding energy eigenvalue of this ground state is simply the zero-point constant $\mathcal{E}_0$:

$$H |\psi_0\rangle = \mathcal{E}_0 |\psi_0\rangle.$$

By acting with the creation operator $\alpha_{\mathbf{k}}^\dagger$ on this vacuum state, we can construct the entire spectrum of excited states:

$$\frac{(\alpha_{\mathbf{k}_1}^\dagger)^{n_{\mathbf{k}_1}}}{\sqrt{n_{\mathbf{k}_1}!}} \frac{(\alpha_{\mathbf{k}_2}^\dagger)^{n_{\mathbf{k}_2}}}{\sqrt{n_{\mathbf{k}_2}!}} \cdots \frac{(\alpha_{\mathbf{k}_N}^\dagger)^{n_{\mathbf{k}_N}}}{\sqrt{n_{\mathbf{k}_N}!}} |\psi_0\rangle,$$

whose corresponding energy eigenvalues are given by:

$$\mathcal{E}_0 + \sum_{i=1}^N n_{\mathbf{k}_i} E_{\mathbf{k}_i},$$

where $n_{\mathbf{k}_i} = \alpha_{\mathbf{k}_i}^\dagger \alpha_{\mathbf{k}_i}$ is the occupation number of the i -th state in the *Bogoliubov quasiparticle basis*.

It is critical to note that the vacuum of these newly defined operators, $|\psi_0\rangle$, is distinct from the vacuum of the original bare particle operators, $a_{\mathbf{k}}$. This distinction becomes physically apparent

when we evaluate the expectation value of the bare particle number operator in the interacting ground state:

$$\begin{aligned}
 \langle \psi_0 | a_{\mathbf{k}}^\dagger a_{\mathbf{k}} | \psi_0 \rangle &= \langle \psi_0 | (u_k \alpha_{\mathbf{k}}^\dagger - v_k \alpha_{-\mathbf{k}}) (u_k \alpha_{\mathbf{k}} - v_k \alpha_{-\mathbf{k}}^\dagger) | \psi_0 \rangle \\
 &= \langle \psi_0 | \left[u_k^2 \alpha_{\mathbf{k}}^\dagger \alpha_{\mathbf{k}} + v_k^2 \alpha_{-\mathbf{k}} \alpha_{-\mathbf{k}}^\dagger - u_k v_k (\alpha_{\mathbf{k}}^\dagger \alpha_{-\mathbf{k}}^\dagger + \alpha_{-\mathbf{k}} \alpha_{\mathbf{k}}) \right] | \psi_0 \rangle \\
 &= v_k^2 \neq 0,
 \end{aligned}$$

where the final evaluation leverages the commutation relation $[\alpha_{-\mathbf{k}}, \alpha_{-\mathbf{k}}^\dagger] = 1$ and the fact that annihilation operators destroy the vacuum. This previous calculation elegantly demonstrates a profound physical consequence of the theory: *even in the absolute zero-temperature ground state $|\psi_0\rangle$, the mean occupation number of states with non-zero momentum is strictly positive.* In terms of the original operators, we would state that $n_{\mathbf{k}} \neq 0$ (where $n_{\mathbf{k}} = a_{\mathbf{k}}^\dagger a_{\mathbf{k}}$) even when $\mathbf{k} \neq 0$. Because of the interactions, a finite fraction of bosons is fundamentally pushed out of the zero-momentum state, leading to a phenomenon known as quantum depletion, where the condensate is only partially occupied even at zero temperature.

1.2 | Condensate Depletion and Excitations

In the previous section, we established that the approximate Hamiltonian of a weakly interacting Bose gas can be rewritten in a diagonal form using the Bogoliubov transformation. The ground state $|\psi_0\rangle$ of this diagonalized Hamiltonian is defined precisely as the vacuum of the new Bogoliubov annihilation operators $\alpha_{\mathbf{k}}$, meaning $\alpha_{\mathbf{k}}|\psi_0\rangle = 0$ for all $\mathbf{k} \neq 0$.

In terms of the original bare particle creation and annihilation operators, we can compute the momentum distribution function in this interacting ground state for both the zero-momentum ($\mathbf{k} = 0$) and non-zero-momentum ($\mathbf{k} \neq 0$) states:

$$\langle\psi_0|a_0^\dagger a_0|\psi_0\rangle = N_0 \sim N,$$

$$\langle\psi_0|a_{\mathbf{k}}^\dagger a_{\mathbf{k}}|\psi_0\rangle = v_k^2 \neq 0, \quad \mathbf{k} \neq 0.$$

While in the non-interacting case all N particles condense into the single-particle state with $\mathbf{k} = 0$, the presence of interactions induces a partial quantum depletion of the condensate even at absolute zero.

We now want to calculate the total number of these non-condensed bosons, denoted as N' . This non-condensate fraction is evaluated by summing the expectation value of the bare number operator over all non-zero momentum states:

$$N' = N - N_0 = \sum'_{\mathbf{k}} \langle\psi_0|a_{\mathbf{k}}^\dagger a_{\mathbf{k}}|\psi_0\rangle = \sum'_{\mathbf{k}} v_k^2,$$

where the primed sum indicates summation over all $\mathbf{k} \neq 0$. As a consistency check, we recall the explicit form of v_k^2 derived earlier; in the strictly non-interacting limit, this excited occupation correctly vanishes:

$$v_k^2 = \frac{1}{2} \left(\frac{\epsilon_k + nU_0}{E_k} - 1 \right) \xrightarrow{U_0 \rightarrow 0} 0.$$

However, for interacting bosons, there is a fundamental partial depletion of the condensate even at zero temperature. Let us calculate the total number of non-condensed bosons N' . To evaluate this macroscopic quantity, we transition to the thermodynamic limit—taking $V \rightarrow \infty$ and $N \rightarrow \infty$ while keeping the density $n = N/V$ constant—which allows us to replace the discrete momentum sum with a continuous integral over k -space:

$$N' = \sum_{\mathbf{k}} v_k^2 \xrightarrow[N, V \rightarrow \infty, n \text{ const.}]{\text{TD limit}} V \int \frac{d^3\mathbf{k}}{(2\pi)^3} v_k^2.$$

Dividing both sides by the volume V , the density of non-condensed bosons n' is given by:

$$n' = \frac{N'}{V} = \int \frac{d^3\mathbf{k}}{(2\pi)^3} v_k^2. \quad (1.2.1)$$

Before explicitly calculating this integral, we will verify its mathematical convergence in both the infrared (long-wavelength, $k \rightarrow 0$) and ultraviolet (short-wavelength, $k \rightarrow \infty$) limits. When $k \rightarrow 0$, the Bogoliubov excitation energy $E_k \rightarrow 0$, which simplifies the occupation number to:

$$v_k^2 \sim \frac{nU_0}{2E_k} \quad (k \rightarrow 0).$$

Infrared limit. To analyze exactly how E_k scales towards zero, we substitute the free particle dispersion $\epsilon_k = \hbar^2 k^2 / 2m$:

$$E_k \sim \sqrt{2nU_0\epsilon_k} = c\hbar k \quad (k \rightarrow 0),$$

where we have conveniently defined the parameter c as:

$$c = \sqrt{\frac{nU_0}{m}}. \quad (1.2.2)$$

This quantity c possesses the dimensions of a velocity and physically corresponds to the *speed of sound* propagating through the weakly interacting Bose gas. Combining these small- k expressions, the ground state occupation number diverges near the origin as:

$$v_k^2 \sim \frac{mc}{2\hbar k} \quad (k \rightarrow 0).$$

Despite this $1/k$ divergence in the integrand, the integration measure $d^3\mathbf{k}$ in spherical coordinates introduces a geometric factor of k^2 , ensuring that the integral (1.2.1) is safely convergent for $k \rightarrow 0$.

Ultraviolet limit. Let us now examine the ultraviolet behavior as $k \rightarrow \infty$. By Taylor expanding the square root definition of E_k in the limit where kinetic energy dominates the interaction ($\epsilon_k \gg nU_0$), we have:

$$E_k = \epsilon_k \sqrt{1 + \frac{2nU_0}{\epsilon_k}} \sim \epsilon_k \left[1 + \frac{nU_0}{\epsilon_k} - \frac{1}{2} \left(\frac{nU_0}{\epsilon_k} \right)^2 \right].$$

Substituting this expansion back into the full expression for v_k^2 , the leading-order terms cancel, leaving a rapid decay:

$$v_k^2 \sim \frac{1}{2} \left[\frac{\epsilon_k + nU_0}{\epsilon_k \left(1 + \frac{nU_0}{\epsilon_k} - \frac{1}{2} \left(\frac{nU_0}{\epsilon_k} \right)^2 \right)} - 1 \right] \sim \frac{(nU_0)^2}{4\epsilon_k^2} = \left(\frac{mc}{\hbar k} \right)^4.$$

Because the occupation number decays as $1/k^4$ at large momenta, the integral converges absolutely at the $k \rightarrow \infty$ upper bound as well.

To formally evaluate the integral, we introduce a natural dimensionless momentum variable $\tilde{k}$ defined as:

$$\tilde{k} = \frac{\hbar k}{mc},$$

such that the occupation number can be elegantly rewritten as a universal function of $\tilde{k}$:

$$v_k^2 = v_{\tilde{k}}^2 = \frac{1}{2} \left(\frac{\tilde{k}^2/2 + 1}{\sqrt{\tilde{k}^4/4 + \tilde{k}^2}} - 1 \right).$$

This final expression is considered universal in the sense that all system-specific dependencies, such as the particle density n or the interaction potential strength U_0 , have been completely absorbed into the definition of the dimensionless coordinate $\tilde{k}$.

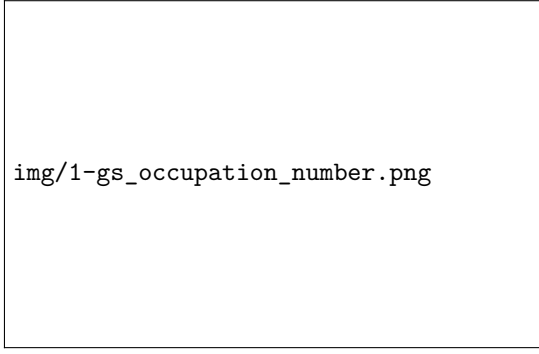


Figure 1.2: Occupation numbers of the ground state v_k^2 . Notice that for small $\tilde{k}$ one has $v_k^2 \sim 1/(2\tilde{k})$, while for large $\tilde{k}$ one has $v_k^2 \sim 1/\tilde{k}^4$.

Having established its convergence and universality, we can explicitly compute the non-condensed density n' by executing the change of variables from k to $\tilde{k}$ inside the integral:

$$n' = \left(\frac{mc}{\hbar}\right)^3 \int \frac{d^3\tilde{k}}{(2\pi)^3} \frac{1}{2} \left(\frac{\tilde{k}^2/2 + 1}{\sqrt{\tilde{k}^4/4 + \tilde{k}^2}} - 1 \right) = \left(\frac{mc}{\hbar}\right)^3 \frac{1}{4\pi^2} \int_0^\infty \tilde{k}^2 \left(\frac{\tilde{k}^2/2 + 1}{\sqrt{\tilde{k}^4/4 + \tilde{k}^2}} - 1 \right) d\tilde{k}.$$

This remaining one-dimensional integral over $\tilde{k}$ can be solved exactly, yielding a factor of $4/3$:

$$\int_0^\infty \tilde{k}^2 \left(\frac{\tilde{k}^2/2 + 1}{\sqrt{\tilde{k}^4/4 + \tilde{k}^2}} - 1 \right) d\tilde{k} = \frac{4}{3}.$$

Applying this result, the density of depleted bosons elegantly simplifies to:

$$n' = \frac{1}{3\pi^2} \left(\frac{mc}{\hbar}\right)^3.$$

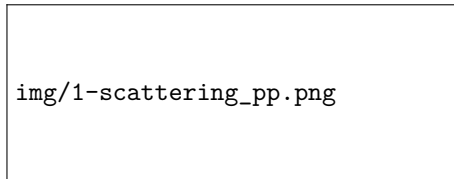
The fundamental fractional depletion of the condensate, given by the ratio $N'/N = n'/n$, is therefore:

$$\frac{N'}{N} = \frac{1}{3\pi^2 n} \left(\frac{mc}{\hbar}\right)^3. \quad (1.2.3)$$

We can thus interpret and manipulate the condensate depletion by operating strictly on the macroscopic density n and the characteristic sound velocity $c = \sqrt{nU_0/m}$. Because c entirely encapsulates the underlying interaction strength U_0 , it effectively acts as the single defining free parameter for the weakly interacting system's thermodynamic properties.

Example (Scattering Length). The scattering length a is a measure of the strength of the interaction between two particles. It is defined as the distance at which the wave function of two interacting particles goes to zero. In the context of Bose-Einstein condensates, the scattering length plays a crucial role in determining the properties of the condensate, including its stability and the nature of its excitations.

Figure 1.3: Scattering process with incoming momentum $\mathbf{p}$ and outgoing momentum $\mathbf{p}'$. If the scattering is elastic, momentum conservation ensures that $|\mathbf{p}'| = |\mathbf{p}|$.



To rigorously justify this parameterization, recall that within the Born approximation—which is formally valid for weak interactions—the scattering amplitude $f(\mathbf{p}' \leftarrow \mathbf{p})$ is directly related

to the Fourier transform of the interaction potential $U(\mathbf{q})$ evaluated at the momentum transfer $\mathbf{q} = \mathbf{p}' - \mathbf{p}$:

$$f(\mathbf{p}' \leftarrow \mathbf{p}) = -\frac{m_r}{2\pi\hbar^2} U(\mathbf{q}),$$

where m_r is the reduced mass of the two-body collision. Because we are considering identical bosons of mass m , the reduced mass simplifies to $m_r = m/2$.

At very low collision energies, the momentum transfer approaches zero, and the scattering becomes completely isotropic, entirely losing its angular dependence. In this fundamental low-energy limit, the scattering length a is mathematically defined as the negative asymptotic limit of the scattering amplitude:

$$a = -\lim_{\mathbf{p}, \mathbf{p}' \rightarrow 0} f(\mathbf{p}' \leftarrow \mathbf{p}).$$

Because the scattering is isotropic, this single parameter is sufficient to completely control both the differential cross-section, $\frac{d\sigma}{d\Omega} = |a|^2$, and the total cross-section, $\sigma_{\text{tot}} = 4\pi a^2$.

By substituting the zero-momentum limit of the Born approximation into this definition, we obtain $a = \frac{m}{4\pi\hbar^2} U_0$. Consequently, for weakly interacting Bose gases, the scattering length can be inverted to define the effective interaction potential U_0 through the relation:

$$U_0 = \frac{4\pi\hbar^2 a}{m}, \quad (1.2.4)$$

where m is the mass of the particles. This relation allows us to express various physical quantities in terms of the scattering length, providing a more intuitive understanding of how interactions affect the condensate.

1.2.1 | Condensate Depletion in Terms of the Gas Parameter

With the effective interaction parameterized strictly by the geometric scattering length a , we can recast our thermodynamic results into a universal form. We first substitute the expression for U_0 into our previous definition of the sound velocity, $c = \sqrt{nU_0/m}$:

$$c = \frac{1}{m} \sqrt{4\pi\hbar^2 n a}.$$

Rearranging these terms to isolate the specific velocity-momentum ratio used in our depletion integral yields:

$$\frac{mc}{\hbar} = 2\sqrt{\pi n a}.$$

We can now insert this expression back into the previously derived equation for the fraction of non-condensed bosons, $\frac{N'}{N} = \frac{1}{3\pi^2 n} \left(\frac{mc}{\hbar}\right)^3$. Performing this substitution directly provides the final analytic result for the condensate depletion:

$$\frac{N'}{N} = \frac{8}{3\sqrt{\pi}} \sqrt{n a^3}. \quad (1.2.5)$$

A profound observation regarding Equation 1.2.5 is the complete absence of the reduced Planck constant, $\hbar$. Because the final result does not contain any $\hbar$ term, there is no mathematical pathway to expand the expression or take a limit to recover a classical or semiclassical analog. This is a direct consequence of the underlying physics: quantum depletion—and Bose-Einstein condensation in general—are purely quantum mechanical effects, fundamentally arising from the zero-point energy and non-commuting nature of the interactions between particles in the condensate.

The quantity N'/N physically represents the fraction of bosons that are pushed out of the condensate ground state. Crucially, Equation 1.2.5 reveals that this depletion depends entirely on a dimensionless quantity called the **gas parameter**, na^3 , which scales as the density multiplied by the characteristic scattering volume. Alternatively, we can interpret this parameter as the cube of the ratio between the scattering length a and the average inter-particle distance $n^{-1/3}$.

For the initial Bogoliubov mean-field treatment to be mathematically valid, we explicitly required the depleted fraction N'/N to be small. Consequently, our theory requires the gas parameter to satisfy the strict condition $na^3 \ll 1$. The real physical meaning of this approximation is that it can be satisfied in two distinct ways: the parameter can be small either because the scattering length a is intrinsically small (representing a genuinely weak interaction), or because the particle density n is exceedingly low (representing a dilute gas).

Because of this density dependence, it can be proven that the Bogoliubov theory remains highly accurate even for infinitely strong interactions, provided the gas is sufficiently dilute. Consider, for instance, a hard-sphere core potential parameterized by a radius r_0 :

$$V(r) = \begin{cases} +\infty & r < r_0, \\ 0 & r > r_0. \end{cases}$$

For such an impenetrable hard-sphere potential, the scattering length is simply equal to the core radius, $a = r_0$. Even in this highly non-perturbative scenario, the established theory remains perfectly valid as long as $nr_0^3 \ll 1$. This specific condition of ultra-low density is universally realized in experimental systems of ultracold atomic gases.

1.3 | Ground State Energy

Let us now study the macroscopic ground state energy $\mathcal{E}_0$ of the interacting system. Recall that for an ideal, non-interacting Bose gas, the ground state energy is trivially zero because all particles occupy the zero-momentum state without any kinetic or potential cost. However, the introduction of a weak interparticle potential fundamentally alters this scenario. By taking the thermodynamic limit, the discrete sum over the momentum modes transforms into a continuous integral over all possible wave vectors, yielding the energy density:

$$\frac{\mathcal{E}_0}{V} = \frac{1}{2}U_0n^2 - \frac{1}{2} \int \frac{d^3k}{(2\pi)^3} (\epsilon_k + nU_0 - E_k).$$

An immediate problem arises when we inspect the behavior of this integral at large momenta (the ultraviolet limit). Based on our earlier asymptotic expansion, the Bogoliubov spectrum E_k approaches the free-particle energy with a leading-order correction, meaning the term in the integrand, $\epsilon_k + nU_0 - E_k$, decays asymptotically as $1/k^2$. Because the three-dimensional integration measure d^3k introduces a k^2 factor, the integral diverges linearly.

The physical origin of this unphysical ultraviolet divergence lies in the strict short-range approximation we previously adopted, namely replacing the momentum-dependent potential $U(\mathbf{k})$ with a constant $U(\mathbf{k} = 0) = U_0$ for all $\mathbf{k}$. In reality, only a mathematically pathological Dirac delta potential possesses a Fourier transform that remains constant for arbitrarily large momenta. For any realistic physical potential, the Fourier transform must eventually decay to zero at large wave vectors.

img/1-physical_potential_FT.png

Figure 1.4: Fourier transform of a generic physical potential $U(\mathbf{r})$ in momentum space, $U(\mathbf{k})$. The Fourier transform is a smooth function that decays to zero at large wave vectors. In contrast, the delta-function approximation for the interaction potential is represented by a constant horizontal line, which is valid only for low-momentum scattering processes.

1.3.1 | Renormalization and the Lee-Huang-Yang Correction

Fortunately, the assumption of a strictly constant potential is not strictly necessary for the theory's structural integrity. If one repeats the rigorous diagonalization procedure while retaining the full momentum dependence of the potential, the Bogoliubov transformation coefficients become:

$$v_{\mathbf{k}}^2 = \frac{1}{2} \left(\frac{\epsilon_k + nU(\mathbf{k})}{E_{\mathbf{k}}} - 1 \right),$$

and the quasiparticle dispersion takes the generalized form:

$$E_{\mathbf{k}} = [(\epsilon_k + nU(\mathbf{k}))^2 - (nU(\mathbf{k}))^2]^{1/2}.$$

With these generalized expressions, the corrected ground state energy density becomes:

$$\frac{\mathcal{E}_0}{V} = \frac{1}{2}U_0n^2 - \frac{1}{2} \int \frac{d^3k}{(2\pi)^3} (\epsilon_k + nU(\mathbf{k}) - E_{\mathbf{k}}). \quad (1.3.1)$$

By evaluating the large momentum limit of this refined integrand, we find:

$$\epsilon_k + nU(\mathbf{k}) - E_{\mathbf{k}} \sim \frac{(nU(\mathbf{k}))^2}{2\epsilon_k}.$$

Therefore, to guarantee the convergence of the energy integral, it is entirely sufficient that the Fourier transform of the potential, $U(\mathbf{k})$, vanishes faster than $1/\sqrt{k}$ at large k . This is a remarkably mild assumption, naturally verified by any real-space potential $U(r)$ that does not diverge faster than $1/r^{5/2}$ as $r \rightarrow 0$.

Although this resolves the mathematical divergence, it introduces an interpretational challenge: the ground state energy now explicitly depends on the high-energy microscopic details of the interaction potential $U(\mathbf{k})$ rather than on a single, universally measurable macroscopic parameter like the scattering length a . To eliminate this unpleasant feature and express the energy purely in terms of physical observables, we employ a standard procedure analogous to classical renormalization. We add and subtract the problematic asymptotic ultraviolet behavior from our integral expression:

$$\frac{\mathcal{E}_0}{V} = \frac{U_0n^2}{2} - \frac{1}{2} \int \frac{d^3k}{(2\pi)^3} \left[\epsilon_k + nU(\mathbf{k}) - E_{\mathbf{k}} - \frac{(nU(\mathbf{k}))^2}{2\epsilon_k} + \frac{(nU(\mathbf{k}))^2}{2\epsilon_k} \right].$$

By splitting the integral and factoring out the density, we obtain:

$$\frac{\mathcal{E}_0}{V} = \frac{n^2}{2} \left(U_0 - \int \frac{d^3k}{(2\pi)^3} \frac{U(\mathbf{k})^2}{2\epsilon_k} \right) - \frac{1}{2} \int \frac{d^3k}{(2\pi)^3} \left[\epsilon_k + nU(\mathbf{k}) - E_{\mathbf{k}} - \frac{(nU(\mathbf{k}))^2}{2\epsilon_k} \right].$$

The bracketed prefactor of the first term can be elegantly linked to the physically observable scattering length a by moving beyond the first-order Born approximation and utilizing the second-order Born approximation for the scattering amplitude $f(\mathbf{p}' \leftarrow \mathbf{p})$:

$$f(\mathbf{p}' \leftarrow \mathbf{p}) = -\frac{m_r}{2\pi\hbar^2} \left[U(\mathbf{p}' - \mathbf{p}) + \int \frac{d^3p''}{(2\pi)^3} \frac{U(\mathbf{p}' - \mathbf{p}'')U(\mathbf{p}'' - \mathbf{p})}{\frac{p^2}{2m_r} - \frac{p''^2}{2m_r} + i0^+} \right],$$

where 0^+ denotes an infinitesimal positive regulator. Recalling the definition of the scattering length in the low-energy limit, the expression simplifies to:

$$a = \frac{m}{4\pi\hbar^2} \left(U_0 - \int \frac{d^3p''}{(2\pi)^3} \frac{U(\mathbf{p}'')^2}{\frac{p''^2}{m} - i0^+} \right).$$

Because the integrand naturally vanishes for $p'' = 0$, the $i0^+$ regulator becomes irrelevant. By changing the integration variable to the wave vector $\mathbf{k}$ and identifying $\epsilon_k = \hbar^2 k^2 / 2m$, we uncover the crucial identity:

$$a = \frac{m}{4\pi\hbar^2} \left(U_0 - \int \frac{d^3k}{(2\pi)^3} \frac{U(\mathbf{k})^2}{2\epsilon_k} \right).$$

By successfully replacing the bare coupling constant U_0 with the physical scattering length a , the ground state energy is renormalized:

$$\frac{\mathcal{E}_0}{V} = \frac{2\pi\hbar^2 n^2 a}{m} - \frac{1}{2} \int \frac{d^3k}{(2\pi)^3} \left(\epsilon_k + nU(\mathbf{k}) - E_{\mathbf{k}} - \frac{(nU(\mathbf{k}))^2}{2\epsilon_k} \right).$$

Because the remaining integral is now manifestly convergent in the ultraviolet regime, we can safely reapply the low-energy approximation $U(\mathbf{k}) \approx U_0$ inside the integral without generating any infinities. The integral then becomes:

$$\int \frac{d^3k}{(2\pi)^3} \left(\epsilon_k + nU_0 - E_k - \frac{(nU_0)^2}{2\epsilon_k} \right).$$

By rewriting this in terms of our previously defined dimensionless momentum variable $\tilde{k}$, the integral scales out as:

$$\left(\frac{mc}{\hbar} \right)^3 mc^2 \frac{1}{2\pi^2} \int_0^\infty \tilde{k}^2 \left(\frac{\tilde{k}^2}{2} + 1 - \sqrt{\frac{\tilde{k}^4}{4} + \tilde{k}^2} - \frac{1}{\tilde{k}^2} \right) d\tilde{k}.$$

This dimensionless integral can be evaluated analytically and yields an exact value of $-32/15$. Inserting this numerical result alongside the definition of the sound velocity c , we arrive at the celebrated final expression for the ground state energy of a dilute Bose gas:

$$\frac{\mathcal{E}_0}{V} = \frac{2\pi\hbar^2 n^2 a}{m} \left[1 + \frac{128}{15\sqrt{\pi}} \sqrt{na^3} \right]. \quad (1.3.2)$$

The leading term in Equation 1.3.2 represents the standard mean-field energy, $\frac{1}{2}n^2U_0$, which precisely matches the expectation value of the Hamiltonian if all quantum field operators were simply replaced by their classical expectation value $\sqrt{n}$. The subsequent correction, proportional to the gas parameter $\sqrt{na^3}$, is the renowned Lee-Huang-Yang (LHY) term. This pivotal term specifically accounts for the zero-point quantum fluctuations of the interacting field around its classical mean-field value.

The macroscopic stability of the gas is dictated by these energy contributions. For a repulsive interaction ($a > 0$), the energy increases quadratically with density, meaning the macroscopic compressibility is positive and the gas will homogeneously fill the confining volume to minimize its energy. Conversely, if the effective potential is attractive ($a < 0$), the scattering length is negative, and the energy is minimized by allowing the density to approach infinity. Because bosons are not protected by the Pauli exclusion principle, an attractive Bose gas is mechanically unstable and will spontaneously undergo spatial collapse.

Remark. *The interplay between the mean-field energy and the LHY quantum fluctuations can yield fascinating new phases of matter. In 2015, D. S. Petrov theoretically demonstrated the stabilization of a collapsing binary Bose-Bose mixture. Consider a mixture of two bosonic species with intra-species repulsion ($a_{11}, a_{22} > 0$) but a sufficiently strong inter-species attraction ($a_{12} < 0$) such that the total mean-field compressibility becomes negative, triggering a collapse. Because the attractive mean-field term scales as $\propto n^2$ while the repulsive LHY fluctuation term scales as $\propto n^{5/2}$, the repulsive quantum fluctuations dominate at higher densities. This delicate competition creates an optimal finite density that minimizes the energy, halting the collapse. The resulting state is a self-bound quantum droplet that remains in equilibrium with the vacuum even without external trapping potentials, a phenomenon successfully observed in experiments in 2018.*

Pressure and Speed of Sound

Returning to a single-component Bose gas, we observe that the macroscopic pressure at absolute zero, $T = 0$, is derived from the standard thermodynamic energy-volume relation:

$$P = - \left(\frac{\partial E}{\partial V} \right)_N = - \frac{E}{V} + n \frac{\partial(E/V)}{\partial n}.$$

By substituting the Lee-Huang-Yang expression for the ground state energy density, we obtain the corrected pressure:

$$P = \frac{2\pi\hbar^2 n^2 a}{m} \left[1 + \frac{64}{5\sqrt{\pi}} \sqrt{na^3} \right].$$

From this thermodynamic equation of state, the macroscopic speed of sound, c_s , in the interacting gas is given by the density derivative of the pressure:

$$c_s^2 = \frac{1}{m} \frac{\partial P}{\partial n} = \frac{4\pi\hbar^2 na}{m^2} \left[1 + \frac{16}{\sqrt{\pi}} \sqrt{na^3} \right].$$

We immediately see that to leading order in the gas parameter, the thermodynamic speed of sound c_s agrees exactly with the microscopic velocity c extracted from the linear slope of the single-particle Bogoliubov dispersion E_k in the long-wavelength limit ($k \rightarrow 0$) introduced previously. It can be formally proven that the fundamental identity $c = c_s$ should hold exactly in any fully consistent many-body theory³⁴. However, maintaining strict consistency with the perturbative result for the speed of sound to next-to-leading order requires extending calculations beyond the present Bogoliubov mean-field approximation when computing the quasiparticle spectrum E_k ⁵.

1.3.2 | Ground State Wave Function

We established that the exact ground state $|\psi_0\rangle$ of the diagonalized effective Hamiltonian must be the vacuum state for the Bogoliubov quasiparticle excitations. Mathematically, this mandates that for all non-zero wave vectors $\mathbf{k} \neq 0$, the annihilation operator yields $a_{\mathbf{k}}|\psi_0\rangle = 0$. If we substitute the defining Bogoliubov transformation—which expresses the quasiparticle operator $\alpha_{\mathbf{k}}$ as a linear combination of the original bare bosonic creation and destruction operators—into this vacuum condition, we obtain:

$$u_k a_{\mathbf{k}} |\psi_0\rangle = -v_k a_{-\mathbf{k}}^\dagger |\psi_0\rangle, \quad (1.3.3)$$

for $\mathbf{k} \neq 0$.

The physical implications of this constraint are profound. On the left-hand side of Equation 1.3.3, the bare annihilation operator $a_{\mathbf{k}}$ acts on the ground state, explicitly removing a boson of momentum $\mathbf{k}$. Simultaneously, on the right-hand side, a creation operator $a_{-\mathbf{k}}^\dagger$ acts on the same ground state, creating a particle with the exact opposite momentum $-\mathbf{k}$. This dynamic equality makes it abundantly clear that the true interacting ground state $|\psi_0\rangle$ cannot be a simple eigenvector of the total particle number operator $\hat{N}$. If it were an eigenstate, the left-hand side of the equation would project onto a subspace with $N - 1$ particles, whereas the right-hand side would project onto a subspace with $N + 1$ particles, creating an irreconcilable contradiction.

³C.R. Cabrera, L. Tanzi, J. Sanz, B. Naylor, P. Thomas, P. Cheiney, and L. Tarruell, *Quantum liquid droplets in a mixture of Bose-Einstein condensates*, Science **359**, 301 (2018).

⁴P. C. Hohenberg and P. C. Martin, Annals of Physics **34**, 291 (1965).

⁵S.T. Beliaev, Soviet Phys. JETP **7**, 299 (1958).

Instead, the structural property demanded by Equation 1.3.3 is satisfied by a state heavily featuring coherent superpositions of correlated particle pairs. We propose the following ansatz for the non-condensate portion of the state:

$$\exp\left(-\frac{1}{2}\sum_{\mathbf{p}}'\frac{v_p}{u_p}a_{\mathbf{p}}^\dagger a_{-\mathbf{p}}^\dagger\right)|0\rangle.$$

To rigorously verify that this state satisfies our requirement, we act upon it with the operator $u_k a_{\mathbf{k}}$:

$$\begin{aligned} u_k a_{\mathbf{k}} \exp\left(-\frac{1}{2}\sum_{\mathbf{p}}'\frac{v_p}{u_p}a_{\mathbf{p}}^\dagger a_{-\mathbf{p}}^\dagger\right)|0\rangle &= u_k \exp\left(-\frac{1}{2}\sum_{\mathbf{p}}'\frac{v_p}{u_p}a_{\mathbf{p}}^\dagger a_{-\mathbf{p}}^\dagger\right)a_{\mathbf{k}}|0\rangle \\ &+ u_k \left[a_{\mathbf{k}}, \exp\left(-\frac{1}{2}\sum_{\mathbf{p}}'\frac{v_p}{u_p}a_{\mathbf{p}}^\dagger a_{-\mathbf{p}}^\dagger\right) \right] |0\rangle. \end{aligned}$$

The first term trivially vanishes because the bare destruction operator $a_{\mathbf{k}}$ acts directly on the absolute vacuum $|0\rangle$. The second term can be explicitly evaluated by expanding the exponential function as a Taylor series. Recalling the fundamental commutator identity $[A, BC] = [A, B]C + B[A, C]$, we evaluate the commutator of a single annihilation operator with a pair creation operator:

$$\begin{aligned} [a_{\mathbf{k}}, a_{\mathbf{p}}^\dagger a_{-\mathbf{p}}^\dagger] &= [a_{\mathbf{k}}, a_{\mathbf{p}}^\dagger]a_{-\mathbf{p}}^\dagger + a_{\mathbf{p}}^\dagger[a_{\mathbf{k}}, a_{-\mathbf{p}}^\dagger] \\ &= a_{-\mathbf{p}}^\dagger \delta_{\mathbf{k}, \mathbf{p}} + a_{\mathbf{p}}^\dagger \delta_{\mathbf{k}, -\mathbf{p}}. \end{aligned}$$

More generally, this can be easily verified by mathematical induction for arbitrary powers of the pair operator:

$$\begin{aligned} [a_{\mathbf{k}}, (a_{\mathbf{p}}^\dagger)^n (a_{-\mathbf{p}}^\dagger)^n] &= [a_{\mathbf{k}}, (a_{\mathbf{p}}^\dagger)^n] (a_{-\mathbf{p}}^\dagger)^n + (a_{\mathbf{p}}^\dagger)^n [a_{\mathbf{k}}, (a_{-\mathbf{p}}^\dagger)^n] \\ &= n(a_{\mathbf{p}}^\dagger)^{n-1} \delta_{\mathbf{k}, \mathbf{p}} (a_{-\mathbf{p}}^\dagger)^n + (a_{\mathbf{p}}^\dagger)^n n(a_{-\mathbf{p}}^\dagger)^{n-1} \delta_{\mathbf{k}, -\mathbf{p}} \\ &= \frac{\partial}{\partial a_{\mathbf{k}}^\dagger} \left((a_{\mathbf{p}}^\dagger)^n (a_{-\mathbf{p}}^\dagger)^n \right). \end{aligned}$$

In this way, one can see that the operational action of the commutator is completely mathematically analogous to a formal differentiation with respect to the conjugate creation operator $a_{\mathbf{k}}^\dagger$. Leveraging this powerful result, we evaluate the commutator with the full exponential:

$$\begin{aligned} \left[a_{\mathbf{k}}, \exp\left(-\frac{1}{2}\sum_{\mathbf{p}}'\frac{v_p}{u_p}a_{\mathbf{p}}^\dagger a_{-\mathbf{p}}^\dagger\right) \right] &= \frac{\partial}{\partial a_{\mathbf{k}}^\dagger} \exp\left(-\frac{1}{2}\sum_{\mathbf{p}}'\frac{v_p}{u_p}a_{\mathbf{p}}^\dagger a_{-\mathbf{p}}^\dagger\right) \\ &= -\frac{1}{2}\sum_{\mathbf{p}}'\frac{v_p}{u_p}(\delta_{\mathbf{k}, \mathbf{p}} a_{-\mathbf{p}}^\dagger + \delta_{\mathbf{k}, -\mathbf{p}} a_{\mathbf{p}}^\dagger) \exp\left(-\frac{1}{2}\sum_{\mathbf{p}}'\frac{v_p}{u_p}a_{\mathbf{p}}^\dagger a_{-\mathbf{p}}^\dagger\right) \\ &= -\frac{v_k}{u_k} a_{-\mathbf{k}}^\dagger \exp\left(-\frac{1}{2}\sum_{\mathbf{p}}'\frac{v_p}{u_p}a_{\mathbf{p}}^\dagger a_{-\mathbf{p}}^\dagger\right). \end{aligned}$$

By substituting this evaluated commutator back into our initial expression, we obtain the exact relation:

$$u_k a_{\mathbf{k}} \exp\left(-\frac{1}{2}\sum_{\mathbf{p}}'\frac{v_p}{u_p}a_{\mathbf{p}}^\dagger a_{-\mathbf{p}}^\dagger\right)|0\rangle = -v_k a_{-\mathbf{k}}^\dagger \exp\left(-\frac{1}{2}\sum_{\mathbf{p}}'\frac{v_p}{u_p}a_{\mathbf{p}}^\dagger a_{-\mathbf{p}}^\dagger\right)|0\rangle,$$

which rigorously proves that our proposed exponential pair-state possesses exactly the property demanded by Equation 1.3.3. Note that the exponential operator acting on the vacuum creates an infinite superposition of bosonic pairs with equal and opposite non-zero momenta, serving as the formal quantum mechanical description of the condensate depletion.

Finally, let us address the macroscopic component corresponding to $\mathbf{k} = 0$. Recalling the mean-field substitution utilized to diagonalize the Hamiltonian ($a_0 \rightarrow \sqrt{N_0}$ and $a_0^\dagger \rightarrow \sqrt{N_0}$), we physically require that the expectation values of the macroscopic creation and annihilation operators over the true ground state equal this condensed value:

$$\langle \psi_0 | a_0 | \psi_0 \rangle = \sqrt{N_0},$$

$$\langle \psi_0 | a_0^\dagger | \psi_0 \rangle = \sqrt{N_0}.$$

These two expectation value equations are simultaneously satisfied by enforcing the eigenvalue relation:

$$a_0 | \psi_0 \rangle = \sqrt{N_0} | \psi_0 \rangle.$$

This defining equation is universally satisfied by Glauber coherent states, which are mathematically structured as:

$$e^{-N_0/2} e^{\sqrt{N_0} a_0^\dagger} |0\rangle.$$

Because the zero-momentum operators completely commute with the pair-creation operators at finite momenta ($\mathbf{p} \neq 0$), both the mean-field condensate requirement and the Bogoliubov pair-correlation requirement can be satisfied simultaneously. Collecting these distinct physical contributions, we arrive at the complete, unified expression for the interacting many-body ground state:

$$| \psi_0 \rangle = \frac{1}{\mathcal{N}} e^{\sqrt{N_0} a_0^\dagger} \exp \left(-\frac{1}{2} \sum_{\mathbf{p}}' \frac{v_p}{u_p} a_{\mathbf{p}}^\dagger a_{-\mathbf{p}}^\dagger \right) |0\rangle, \quad (1.3.4)$$

where $\mathcal{N}$ is an overall normalization constant that ensures $\langle \psi_0 | \psi_0 \rangle = 1$. By explicitly computing the norm of this state, this constant can be analytically proven to be:

$$\mathcal{N} = e^{N_0/2} \left(\prod_{\mathbf{k}} u_k^2 \right)^{1/2}.$$

1.4 | Spontaneous Symmetry Breaking and Goldstone Theorem

It is straightforward to verify that the original unperturbed Hamiltonian commutes with the total number operator $\hat{N} = \sum_{\mathbf{k}} a_{\mathbf{k}}^\dagger a_{\mathbf{k}}$. This commutation property is intimately connected with the fundamental invariance of the Hamiltonian under global U(1) phase transformations:

$$a_{\mathbf{k}} \rightarrow e^{i\phi} a_{\mathbf{k}},$$

$$a_{\mathbf{k}}^\dagger \rightarrow e^{-i\phi} a_{\mathbf{k}}^\dagger.$$

These phase rotations are formally realized by the application of the unitary continuous symmetry operator $U(\phi)$:

$$U(\phi) = e^{i\phi\hat{N}}.$$

Indeed, it is easy to explicitly check the generator action on the annihilation operator:

$$U^\dagger(\phi) a_{\mathbf{k}} U(\phi) = e^{i\phi} a_{\mathbf{k}}.$$

Because the Hamiltonian is invariant under this specific transformation—meaning $U^\dagger(\phi) H U(\phi) = H$ —Noether’s theorem guarantees there should be a conserved macroscopic charge. This conserved charge is nothing else than the mathematical generator of the transformation itself, namely, the total number operator $\hat{N}$.

Crucially, even if the Hamiltonian governing the system is perfectly invariant under the above U(1) transformation, the interacting ground state $|\psi_0\rangle$ is not. This profound physical phenomenon is known as spontaneous symmetry breaking. There are multiple analytical ways to see this symmetry breaking. Consider the macroscopic expectation value of the zero-momentum annihilation operator a_0 . We previously established that over the coherent ground state, $\langle\psi_0|a_0|\psi_0\rangle \neq 0$. Applying the unitary transformation, we have:

$$\langle\psi_0|(U^\dagger(\phi)a_0U(\phi))|\psi_0\rangle = \langle\psi_0|(e^{i\phi}a_0)|\psi_0\rangle.$$

However, if the ground state $|\psi_0\rangle$ were truly invariant under the symmetry group, then by definition $U(\phi)|\psi_0\rangle = |\psi_0\rangle$. This invariant assumption would require that the equation evaluates to:

$$\langle\psi_0|a_0|\psi_0\rangle = e^{i\phi}\langle\psi_0|a_0|\psi_0\rangle,$$

which can logically be satisfied for an arbitrary phase ϕ if and only if $\langle\psi_0|a_0|\psi_0\rangle = 0$. This immediately leads to a direct contradiction with our foundational result that the condensate is macroscopically occupied. The exact same mathematical proof can be repeated for the pair correlation expectation value at finite momenta $\mathbf{k} \neq 0$. In that case, we notice that

$$\langle\psi_0|a_{\mathbf{k}}a_{-\mathbf{k}}|\psi_0\rangle = -u_k v_k \neq 0,$$

and applying the operator $U(\phi)$ to both bare operators and falsely imposing the invariance of the ground state $|\psi_0\rangle$ once again yields a fatal contradiction.

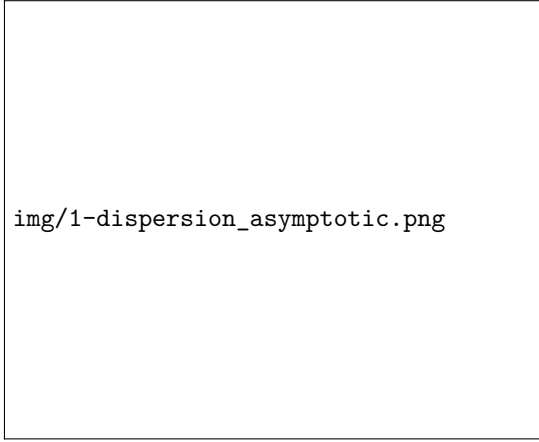
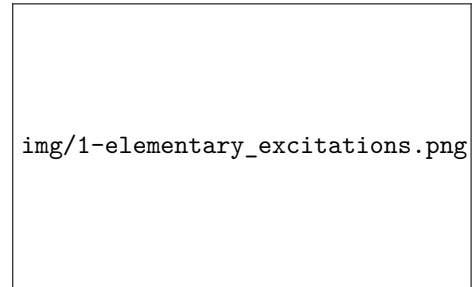


Figure 1.5: Plot of the dispersion $E_k = \sqrt{\epsilon_k^2 + 2nU_0\epsilon_k}$ (blue line) with its two asymptotic behaviours: for $k \rightarrow 0$ we have that the dispersion is linear with $E_k \sim chk$, while for $k \rightarrow \infty$ we have that the dispersion is quadratic with $E_k \sim \epsilon_k + nU_0$.

Once we have rigorously verified that the chosen ground state explicitly breaks the continuous $U(1)$ symmetry, we know by Goldstone's theorem that there must emerge massless Goldstone bosons. These excitations must exhibit an energy dispersion that vanishes continuously (specifically, linearly) with the momentum p , yielding $E \sim cp$ for $p \rightarrow 0$. Recall that after executing the Bogoliubov transformation, the interacting Hamiltonian becomes diagonalized, featuring the quasiparticle energy spectrum given by E_k . The asymptotic limit $k \rightarrow 0$ for E_k has been already evaluated, successfully reproducing exactly the linear phononic dispersion that Goldstone's theorem demands: $E_k \sim chk$ (as visualized in Figure 1.5), with the characteristic sound velocity c scaling with the interaction strength.

Figure 1.6: Elementary excitations are modes that allow us to describe low energy states. If we excite the ground state GS, increasing the energy by $+E_k$, it's like creating an elementary excitation (a quasi-particle).



The derived energy values E_k structurally describe the dispersion of these elementary excitations, which are collective modes that allow us to describe effectively low-energy many-body states as a simple ideal gas of quasi-particles (see Figure 1.6). This quasiparticle framework is particularly relevant at very low temperatures T , when only these low-energy excited states are thermally activated, since random thermal fluctuations are small and strictly of order $k_B T$.

Consequently, complex interacting states at low energy can be cleanly described as a dilute gas of elementary excitations. It is critical to recognize that these independent excitations are not the original bare bosons created by the standard operators $a_{\mathbf{k}}^\dagger$, but rather new hybridized bosonic modes created by the Bogoliubov operators $\alpha_{\mathbf{k}}^\dagger$ for small k . These elementary excitations are governed by an effective interacting dispersion E_k that is structurally distinct from the purely quadratic free-particle dispersion ϵ_k of the original bare bosons.

Remarkably, even for a strongly interacting bare Bose gas, provided the temperature remains sufficiently low, very few of these hybridized modes are activated. Because the thermal density of quasi-particles is low, these elementary excitations are highly dilute and can thus be treated to an excellent approximation as a non-interacting ideal gas. Lastly, we note a fundamental statistical

constraint: for a system comprised exclusively of underlying bosons, like in the present model, only bosonic elementary excitations are physically possible. In contrast, for a many-body system of underlying fermions, the collective excitations can manifest as either fermionic (quasiparticles) or bosonic (e.g., Cooper pairs, plasmons) elementary excitations depending on the specific pairing mechanics.

1.5 | Specific Heat and Condensate Fraction

Let us proceed with the calculation of the specific heat of a dilute Bose gas. From the diagonalized Hamiltonian, we established that at absolute zero temperature, $T = 0$, the system resides in the quasiparticle vacuum with no elementary excitations. At a finite temperature $T > 0$, the thermal mean number of elementary excitations, $\langle \alpha_{\mathbf{k}}^\dagger \alpha_{\mathbf{k}} \rangle$, is governed by the standard Bose-Einstein distribution—since these Bogoliubov quasiparticles are fundamentally bosons:

$$\langle \alpha_{\mathbf{k}}^\dagger \alpha_{\mathbf{k}} \rangle = \frac{1}{e^{\beta E_k} - 1},$$

where $\beta = (k_B T)^{-1}$. In this expression, the chemical potential μ associated with the elementary excitations is strictly zero. This physically arises because the total number of quasiparticle excitations is not a conserved quantity; it is determined dynamically by the thermal equilibrium at temperature T (entirely analogous to phonons in a crystalline solid or photons in black-body radiation). Consequently, the total internal energy $E(T)$ of the gas at temperature T is given by the sum of the zero-point ground state energy and the thermal excitation energy:

$$E(T) = \mathcal{E}_0 + \sum_{\mathbf{k}} \frac{E_k}{e^{\beta E_k} - 1}.$$

Taking the thermodynamic limit, the discrete summation over wave vectors transitions into a continuous integral, yielding the total energy density:

$$\frac{E(T)}{V} = \frac{\mathcal{E}_0}{V} + \int \frac{d^3 k}{(2\pi)^3} \frac{E_k}{e^{\beta E_k} - 1}.$$

For high-energy modes where $E_k \gtrsim k_B T$, the integrand is exponentially suppressed by the Bose distribution. Therefore, the dominant contribution to the integral comes strictly from the low-energy domain where $E_k \lesssim k_B T$. At very low temperatures, this restricts our focus to the extreme long-wavelength limit of the dispersion relation.

Figure 1.7: At low temperature T , if we are interested just in the region $E_k \lesssim k_B T$, we can see that the dispersion E_k goes linearly with the momentum k .

TODO: add image of a graph showing the quasiparticle dispersion energy curve plotted against momentum, with a highlighted circle near the origin emphasizing the linear regime at low energies comparable to thermal fluctuations

In this low-momentum regime, we recall that the Bogoliubov dispersion is strictly linear, $E_k \approx c\hbar k$. Thus, in the asymptotic limit $T \rightarrow 0$, we can safely replace the full dispersion relation with its phononic approximation inside the integral:

$$\frac{E(T)}{V} = \frac{\mathcal{E}_0}{V} + \int \frac{d^3 k}{(2\pi)^3} \frac{c\hbar k}{e^{\beta c\hbar k} - 1}.$$

Evaluating this integral by switching to spherical coordinates and introducing the dimensionless integration variable $x = \beta c\hbar k$, we obtain:

$$\begin{aligned} \frac{E(T)}{V} &= \frac{\mathcal{E}_0}{V} + \left(\frac{k_B T}{c\hbar} \right)^3 k_B T \frac{4\pi}{(2\pi)^3} \int_0^\infty \frac{x^3}{e^x - 1} dx \\ &= \frac{\mathcal{E}_0}{V} + \frac{\pi^2}{30} \frac{(k_B T)^4}{(c\hbar)^3}. \end{aligned}$$

From this energy density, we directly compute the isochoric heat capacity, C_V :

$$C_V = \left(\frac{\partial E(T)}{\partial T} \right)_V = \frac{2\pi^2}{15} V k_B \left(\frac{k_B T}{c\hbar} \right)^3.$$

Figure 1.8: Plot of the behavior of the specific heat C_V at low temperature for a bosonic system. The blue line represents the specific heat of an interacting Bose gas, which scales proportionally to the cube of the temperature. The red line represents the specific heat of a non-interacting Bose gas, scaling proportionally to the temperature to the power of three halves.

This final expression reveals that the low-temperature specific heat of the weakly interacting Bose gas scales as T^3 . This characteristic cubic temperature dependence mirrors the behavior of acoustic phonons in a Debye solid or photons in a cavity, fundamentally stemming from the linear phononic dispersion relation. We can starkly contrast this interacting result with the ideal, non-interacting Bose gas, which exhibits a $T^{3/2}$ dependence due to its purely quadratic free-particle dispersion. In the interacting system, the linear dispersion creates a lower density of states at low energies compared to the non-interacting case, making the elementary excitations significantly harder to thermally excite at temperatures near absolute zero.

TODO: add image of a graph comparing two curves of specific heat against temperature, showing a cubic blue curve for the interacting gas and a three halves power red curve for the ideal gas

1.6 | Landau Criterion for Superfluidity

[...]

Exercises

Sheet 1

[...]

Appendices

A | Notation and Conventions

B | Computations and further developments